

6(a)	product of (magnetic) flux density and area	M1
	where area is perpendicular to the (magnetic) field	A1
6(b)(i)	$N\Phi = BAN$	C1
	$= 400 \times 10^{-3} \times 0.12^2 \times 8$	C1
	$= 0.046 \text{ Wb}$	A1
6(b)(ii)	(line is a) straight line	B1
6(b)(iii)	(induced) e.m.f. = rate of change of flux linkage	C1
	e.m.f. $= N\Phi / t$	A1
	$= 0.046 / 0.60$	
	$= 0.077 \text{ V}$	
6(c)	(induced e.m.f. causes) current flow (in the coil)	B1
	either	
	current (in magnetic field) causes forces to act on the coil	B1
	(opposite sides of) coil forced inwards	B1
	or	
	current causes dissipation of energy in the resistance of the coil	(B1)
	temperature of the coil rises	(B1)